Quantum information and computation

Single and multiple systems

October 11, 2026

Outline

1. Single systems
2. Multiple systems

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1. Single systems
   - Classical information
   - Quantum information
2. Multiple systems

Introduction

- Consider X a physical system that stores information.
- Assume X can be in one of a finite number of classical states, $\mathcal{S}$, at each moment. For instance:
  - If X is a bit, then its classical state set is $\mathcal{S} = \{0, 1\}$.
  - If X is a six-sided die, then $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$.
  - If X is a switch on a standard electric fan, then perhaps $\mathcal{S} = \{\text{high}, \text{ medium}, \text{ low}, \text{ off}\}$.
- There may be uncertainty about the classical state of a system, where each classical state has some probability associated with it.

Introduction

- We may not know the exact state, but we know the probability associated with each possible outcome.
- **Exampe 1**: consider a coin with two possible states: $\mathcal{S} = \{\text{Heads}, \text{Tails}\}$.
  - If the coin is biased, the probabilities may be $P(X = \text{Heads}) = 0.3, \quad P(X = \text{Tails}) = 0.7$
- **Example 2**, if X is a bit,
  - It might be in the classical state 0 with probability $\frac{3}{4}$ and in the classical state 1 with probability $\frac{1}{4}$.
  - This probabilistic state could be represented by a column vector:

$$\begin{pmatrix} \frac{3}{4} \\ \frac{1}{4} \end{pmatrix} \quad \begin{matrix} \leftarrow \text{entry corresponding to 0} \\ \leftarrow \text{entry corresponding to 1} \end{matrix}$$

- All entries are nonnegative real numbers, their sum is 1.

Dirac notation, First part

- Let $\mathcal{S}$ be a classical state set, with its elements labeled as $1, \ldots, |\mathcal{S}|$.
- For each $a \in \mathcal{S}$, we define the column vector $|a\rangle$ to have a 1 in the position corresponding to a, and 0 in all other positions.
- **Example 1**: let's $\mathcal{S} = \{0, 1\}$, the basis vectors are

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

- **Example 2**: let's $\mathcal{S} = \{\clubsuit, \diamondsuit, \heartsuit, \spadesuit\}$, the basis vectors are:

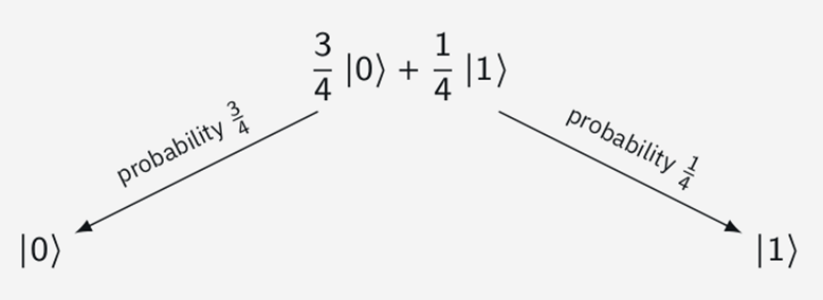
$$|\clubsuit\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad |\diamondsuit\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad |\heartsuit\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad |\spadesuit\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

Dirac notation, First part

- Vectors of this form are called *standard basis vectors*.
- Every vector can be expressed uniquely as a linear combination of standard basis vectors.

$$\begin{pmatrix} \frac{3}{4} \\ \frac{1}{4} \end{pmatrix} = \frac{3}{4}|0\rangle + \frac{1}{4}|1\rangle$$

- Measuring the system selects (or reveals) a transition, chosen at random:



Deterministic operations

- Every function $f : \mathcal{S} \to \mathcal{S}$ represents a **deterministic operation** that maps a classical state a to the state $f(a)$ for each $a \in \mathcal{S}$.
- For any function $f : \mathcal{S} \to \mathcal{S}$, there exists a unique matrix M such that

$$M|a\rangle = |f(a)\rangle \quad \text{for all } a \in \mathcal{S}.$$

- The matrix M contains exactly one entry equal to 1 in each column, while all other entries are 0. More precisely,

$$M(b,a) = \begin{cases} 1 & \text{if } b = f(a), \\ 0 & \text{otherwise.} \end{cases}$$

- The effect of this operation on a vector v is then given by the **matrix–vector multiplication**

$$v \mapsto Mv.$$

Deterministic operations

- For $\mathcal{S} = \{0, 1\}$, there exist four fonctions $f : \mathcal{S} \to \mathcal{S}$:

a	$f_1(a)$
0	0
1	0

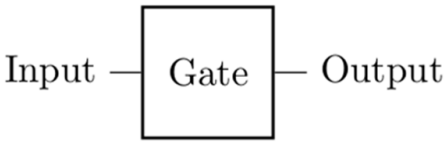
a	$f_2(a)$
0	0
1	1

a	$f_3(a)$
0	1
1	0

a	$f_4(a)$
0	1
1	1

- Here are their corresponding matrices:

$$M_1 = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \quad M_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad M_3 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad M_4 = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$$



Input	Output
0	?
1	?

Dirac notation, Second part

- Let $\mathcal{X}$ be any classical state set, and assume the elements of $\mathcal{X}$ have been placed in correspondence with the integers $1, \ldots, |\mathcal{X}|$.
- We denote by $\langle a|$ the row vector having a 1 in the entry corresponding to $a \in \mathcal{X}$, with 0 for all other entries.
- Let $\mathcal{X} = \{0, 1\}$. The standard basis vectors are (**kets**):

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

- Their corresponding row vectors (**bras**) are:

$$\langle 0| = \begin{pmatrix} 1 & 0 \end{pmatrix}, \quad \langle 1| = \begin{pmatrix} 0 & 1 \end{pmatrix}$$

Dirac notation, Second part

- Multiplying a row vector to a column vector yields a scalar:

$$\begin{pmatrix} * & * & * & \cdots \end{pmatrix} \begin{pmatrix} * \\ * \\ * \\ \vdots \end{pmatrix} \qquad \langle a \mid b \rangle = \begin{cases} 1 & \text{if } a = b \\ 0 & \text{if } a \neq b \end{cases}$$

- Multiplying a column vector to a row vector yields a matrix:

$$\begin{pmatrix} * \\ * \\ * \\ \vdots \\ * \end{pmatrix} \begin{pmatrix} * & * & * & \cdots & * \end{pmatrix} = \begin{pmatrix} ** & ** & ** & \cdots & ** \\ ** & ** & ** & \cdots & ** \\ ** & ** & ** & \cdots & ** \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ ** & ** & ** & \cdots & ** \end{pmatrix}$$

$$|0\rangle\langle 0| = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

Dirac notation, Second part

- Given a function $f : \mathcal{S} \to \mathcal{S}$, the corresponding matrix M can be written as

$$M = \sum_{b \in \mathcal{S}} |f(b)\rangle\langle b|.$$

- Its action on a standard basis vector $|a\rangle$ is:

$$M|a\rangle = \left(\sum_{b \in \mathcal{S}} |f(b)\rangle\langle b| \right) |a\rangle = \sum_{b \in \mathcal{S}} |f(b)\rangle\langle b|a\rangle = |f(a)\rangle,$$

- Example: let $\mathcal{S} = \{0, 1\}$ and $f(0) = 1, f(1) = 0$. Then

$$M = |f(0)\rangle\langle 0| + |f(1)\rangle\langle 1| = |1\rangle\langle 0| + |0\rangle\langle 1| = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Introduction

- A quantum state of a system is described by a column vector, where each index corresponds to one of the system's classical states:
  - Each entry of the vector is a complex number.
  - The sum of the squares of the absolute values of the entries equals 1.
- Quantum state vectors are therefore unit vectors with respect to their Euclidean norm, with complex number entries α_k:

$$v = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix} \quad \Longrightarrow \quad \|v\| = \sqrt{\sum_{k=1}^{n} |\alpha_k|^2}.$$

Dirac notation, third part

- Example 1: $|0\rangle$ and $|1\rangle$

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

- Example 2: $\{\clubsuit, \diamondsuit, \heartsuit, \spadesuit\}$:

$$|\psi\rangle = \frac{1}{2}|\clubsuit\rangle - \frac{i}{2}|\diamondsuit\rangle + 0 \cdot |\heartsuit\rangle + \frac{\sqrt{2}}{2}|\spadesuit\rangle = \begin{pmatrix} \frac{1}{2} \\ -\frac{i}{2} \\ 0 \\ \frac{\sqrt{2}}{2} \end{pmatrix}$$

- The notation $|\psi\rangle$ is commonly used to refer to an arbitrary quantum state:

$$|\psi\rangle = \frac{1+2i}{2}|0\rangle - |1\rangle$$

- For any column vector $|\psi\rangle$, the row vector $\langle\psi|$ is its conjugate transpose, $\langle\psi| = (|\psi\rangle)^\dagger$.

Measuring quantum states

- The probability for each classical state to be the outcome is the absolute value squared of the corresponding quantum state vector entry.

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

- Measuring $|+\rangle$ yields:

$$\Pr(\text{outcome } 0) = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}, \quad \Pr(\text{outcome } 1) = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$$

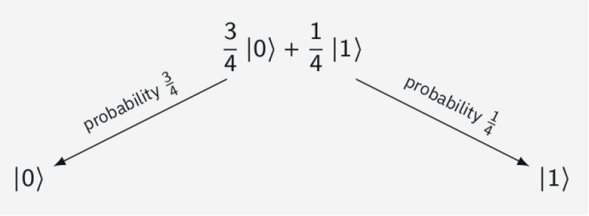
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Unitary operations

- The set of allowable operations on quantum states differs from classical information comes from the fundamental rules of quantum mechanics, which impose stricter constraints than classical physics:
  - Quantum operations must be linear and unitary.
  - Quantum states can exist in superpositions.
  - Quantum states of multiple qubits can be entangled, meaning the state of one qubit depends on the others.
  - Measuring a quantum state collapses the superposition probabilistically.

Unitary operations

- Square matrix U with complex entries is **unitary** if it satisfies, where $U^\dagger$ is the conjugate transpose of U, and I is the identity matrix.

$$U^\dagger U = I = UU^\dagger$$

  - $U^\dagger$ means first taking the transpose of U and then taking the complex conjugate of each entry.
  - Unitarity ensures that U preserves the Euclidean norm of any vector it acts on, which is why all quantum operations must be unitary (except measurement).

Unitary operations

- If U is a unitary matrix, then for any vector v, where $\|v\|$ is the Euclidean (or 2-) norm of v, defined as $\|v\| = \sqrt{\sum_i |v_i|^2}$

$$\|Uv\| = \|v\|$$

- Example:

$$v = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

- Applying U to v and then compute the Euclidean norms:

$$\|Uv\| = \sqrt{|v_0'|^2 + |v_1'|^2} = \sqrt{\frac{1}{2} + \frac{1}{2}} = 1 = \|v\| \rightarrow (\sqrt{1^2 + 0^2})$$

Unitary operations: Pauli

- Pauli operations are ones represented by the Pauli matrices:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

- Common alternative notations: $X = \sigma_x$, $Y = \sigma_y$, $Z = \sigma_z$.

- The operation σ_x is also called a bit flip (or NOT operation), and σ_z is called a phase flip:

$$\sigma_x|0\rangle = |1\rangle, \quad \sigma_x|1\rangle = |0\rangle, \quad \sigma_z|0\rangle = |0\rangle, \quad \sigma_z|1\rangle = -|1\rangle.$$

Unitary operations: Hadamard

- The Hadamard operation is represented by this matrix:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

- H is unitary because $H^\dagger H = I$, the identity matrix.

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad H^\dagger = H^T = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$H^\dagger H = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1+1 & 1-1 \\ 1-1 & 1+1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

Unitary operations: Phase

- A phase operation is one described by the matrix.

$$P_\theta = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix},$$

- For any real number θ, two important special cases are the S and T gates:

$$S = P_{\pi/2} = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \qquad T = P_{\pi/4} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$$

- P_θ is the phase gate that adds a phase $e^{i\theta}$ to the $|1\rangle$ state.

$$P_\theta|1\rangle = e^{i\theta}|1\rangle.$$

Unitary operations: Exercice

- Verify the following relations by performing the matrix multiplications:

$$H|0\rangle = |+\rangle, \qquad H|1\rangle = |-\rangle,$$

$$H|+\rangle = |0\rangle, \qquad H|-\rangle = |1\rangle.$$

- Show that the T gate acts on the computational basis as

$$T|0\rangle = |0\rangle, \qquad T|1\rangle = \frac{1+i}{\sqrt{2}}|1\rangle.$$

- Let $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. Compute $T|+\rangle$ using linearity.

Composing unitary operations

- Compositions of unitary operations are represented by matrix multiplication.
- Applying a Hadamard operation, followed by the phase operation S, followed by another Hadamard yields:

$$HSH = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}\begin{pmatrix}1 & 0\\ 0 & i\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix} = \frac{1}{2}\begin{pmatrix}1+i & 1-i\\ 1-i & 1+i\end{pmatrix}$$

Applying this unitary operation twice yields a NOT (Pauli-X) operation:

$$(HSH)^2 = \left(\frac{1}{2}\begin{pmatrix}1+i & 1-i\\ 1-i & 1+i\end{pmatrix}\right)^2 = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}$$

Outline

1 Single systems

2 Multiple systems
- Classical states
- Quantum states

Definitions

- Suppose we have two **classical systems**:
  - X, with a set of possible states A;
  - Y, with a set of possible states B.
- If we place X and Y side by side, with X on the left and Y on the right, we can view them together as a single **compound system**.
- We denote this compound system by (X, Y) or simply XY. The state of the compound system is then a *pair* (x, y) where $x \in A$ and $y \in B$.
- The set of all possible states of the compound system is the **Cartesian product** of the individual state sets, denoted by C:

$$C = A \times B = \{(x, y) \mid x \in A,\ y \in B\}.$$

Cartesian product

- Example 1
  - A and B are the states of the first and second system. C is the Cartesian product of A and B

$$A = \{0, 1\} \quad \text{and} \quad B = \{\clubsuit, \diamondsuit, \heartsuit, \spadesuit\},$$

  - The Cartesian product of A and B is:

$$C = A \times B = \{(0, \clubsuit), (0, \diamondsuit), (0, \heartsuit), (0, \spadesuit), (1, \clubsuit), (1, \diamondsuit), (1, \heartsuit), (1, \spadesuit)\}.$$

- Example 2
  - X_1, X_2, and X_3 with state sets $A = B = D = \{0, 1\}$.
  - The compound system (X_1, X_2, X_3) is ($|00\rangle$ notation):

$$C = A \times B \times D = \left\{ \begin{matrix} (0,0,0), (0,0,1), (0,1,0), (0,1,1), \\ (1,0,0), (1,0,1), (1,1,0), (1,1,1) \end{matrix} \right\}.$$

Definitions

- **Quantum state** vectors of **composite systems** are represented as column vectors, with indices corresponding to the **Cartesian product** of the individual systems' classical state sets.
  - If X and Y are qubits, the classical state set for the pair (X, Y) is

$$\{0, 1\} \times \{0, 1\} = \{00, 01, 10, 11\}.$$

  - These are examples of quantum state vectors of the pair (X, Y):

$$\frac{1}{2}\Big(|00\rangle + |01\rangle - |10\rangle - |11\rangle\Big), \qquad \frac{1}{\sqrt{2}}\Big(|00\rangle - |11\rangle\Big).$$

Tensor products

- Tensor products of quantum state vectors are also quantum state vectors.
- Let $|\phi\rangle$ be a quantum state vector of a system X, and let $|\psi\rangle$ be a quantum state vector of a system Y. Then the tensor product, $|\phi\rangle \otimes |\psi\rangle$ is a quantum state vector of the compound system (X, Y).
- States of this form are called *product states*, and they represent independence between the systems X and Y.
- More generally, if $|\psi_1\rangle, \ldots, |\psi_n\rangle$ are quantum state vectors of systems $X_1, \ldots, X_n$, then $|\psi_1\rangle \otimes \cdots \otimes |\psi_n\rangle$ is a quantum state vector representing a product state of the compound system $(X_1, \ldots, X_n)$.

Tensor products

- The quantum state vector, an example of a *product state*

$$\frac{1}{2}\Big(|00\rangle + |01\rangle - |10\rangle - |11\rangle\Big)$$

- It can be written as a tensor product of single-qubit states:

$$\frac{1}{2}\Big(|00\rangle + |01\rangle - |10\rangle - |11\rangle\Big) = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle).$$

Tensor products

- Exercice: Why the quantum state vector of two qubits is *not* a product state.

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

- It is an example of a quantum state vector, one of the four Bell states, which collectively form the Bell basis.

$$|\phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \qquad |\phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle),$$

$$|\psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle), \qquad |\psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle).$$

Measurements

- Measurements of compound systems work in the same way as measurements of single systems-provided that all of the subsystems are measured.
- Example, the pair (X, Y) is in the quantum state

$$|\psi\rangle = \frac{3}{5}i\,|0\rangle|\heartsuit\rangle - \frac{4}{5}\,|1\rangle|\spadesuit\rangle,$$

- Measuring both systems yields one of the outcomes $(0, \heartsuit)$ or $(1, \spadesuit)$, with probabilities determined by the squared magnitudes of the coefficients:

$$P(0, \heartsuit) = \left|\frac{3}{5}i\right|^2 = \frac{9}{25}, \qquad P(1, \spadesuit) = \left|-\frac{4}{5}\right|^2 = \frac{16}{25}.$$

Measurements

- Two systems (X, Y) are together in some quantum state. What happens when we measure X and do nothing to Y?
- A quantum state vector of (X, Y) can be written as follows, where $a \in A$ and $b \in B$ are the classical states of X and Y, respectively.

$$|\psi\rangle = \sum_{(a,b)\in A\times B} \alpha_{ab} |ab\rangle,$$

- If both X and Y are measured, then each outcome $(a, b) \in A \times B$ appears with probability.

$$P(a, b) = |\langle ab|\psi\rangle|^2 = |\alpha_{ab}|^2.$$

- α_{ab} is the coefficient in the quantum state corresponding to the basis state $|ab\rangle$

Unitary operations

- Quantum operations on compound systems are represented by unitary matrices whose rows and columns correspond to the Cartesian product of the classical state sets of the individual systems.
- The combined action of a collection of unitary operations applied independently to a collection of systems is represented by the tensor product of the unitary matrices.
  - If $X_1, \ldots, X_n$ are quantum systems and $U_1, \ldots, U_n$ are unitary operations applied independently to them, the combined operation on $(X_1, \ldots, X_n)$ is $U_1 \otimes \cdots \otimes U_n$.
  - Performing a unitary U on system X while leaving Y unchanged is represented by $U \otimes I_Y$, where I_Y is the identity on Y.

Unitary operations

- Suppose X and Y are qubits. Performing a Hadamard operation on X and doing nothing to Y is represented by the unitary operation

$$H \otimes I = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix}$$

- Performing a Hadamard operation on Y while leaving X unchanged is represented by

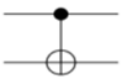
$$I \otimes H = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \end{pmatrix}$$

Unitary operations

Pauli-X (X)		$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Pauli-Y (Y)		$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
Pauli-Z (Z)		$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Hadamard (H)		$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
Phase (S, P)		$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$
$\pi/8$ (T)		$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$

Unitary operations

Controlled Not (CNOT, CX)



$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Controlled Z (CZ)



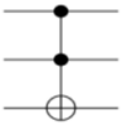
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

SWAP



$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Toffoli (CCNOT, CCX, TOFF)



$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

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